

Restaurant Management System

Software Requirements Specification

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1 Introduction

This document is the Software Requirements Specification (SRS) for the Restaurant Management System developed by Team Zenith. It describes the functional and non-functional requirements of the system, defines the scope of the project, identifies the users of the system, and outlines the constraints and assumptions under which the system will operate.

The system is designed for small independent restaurants and serves as both an internal management tool for staff and administrators, and a customer-facing ordering interface accessible via table QR codes. This document serves as the reference point for all development decisions and is intended to be read alongside the System Design Document.

2 Purpose

The purpose of this system is to replace manual, paper-based restaurant operations with a centralized digital platform built specifically for small independent restaurants that cannot afford enterprise-level solutions.

The Restaurant Management System addresses this by providing:

- A digital menu with real-time availability and archiving managed by authorized staff
- A QR-based table ordering system allowing customers to place orders from their own devices
- An internal order management system with item-level tracking and kitchen display
- Automated wait time estimation based on active kitchen load
- A visual drag-and-drop staff scheduling system with shift switch requests
- A public surplus food distribution forum to reduce food waste and support the local community
- A feedback and ratings system to give restaurant owners actionable customer insight

3 Scope

3.1 What the System Does

- Provides a QR-based ordering interface for customers linked to their specific table
- Manages the full order lifecycle from placement through preparation to delivery and billing
- Tracks kitchen load in real time and estimates wait times per order
- Gives staff a dedicated kitchen display with item-level status controls
- Allows admin to manage menu items, categories, tables, and user accounts

- Provides a visual timetable for staff scheduling with drag and drop shift management
- Handles shift switch requests between staff members with admin approval
- Generates bills automatically and supports UPI QR payment on the customer's device
- Publishes surplus food availability to a public page/forum at end of day
- Collects optional post-meal customer ratings

3.2 What the System Does Not Do

- Does not process payments directly — UPI QR is displayed and payment is confirmed externally
- Does not support online delivery or takeaway tracking
- Does not track ingredient-level inventory or stock
- Does not provide a native mobile application

4 Definitions and Abbreviations

Term	Definition
Admin	Authenticated user with full system access including scheduling and user management
Staff	Authenticated user with operational access to orders, menu availability, and scheduling
Customer	Unauthenticated user who accesses the system via table QR code and PIN
RLS	Row Level Security — database-level access control enforced by Supabase
JWT	JSON Web Token — session token issued on login
UUID	Universally Unique Identifier — used as primary keys in all database tables
CRUD	Create, Read, Update, Delete
API	Application Programming Interface
QR	Quick Response code — used for table-based customer access
PIN	4-digit code given to customers by staff to access their table session
Supabase Realtime	Live database subscription feature used for real-time order and table updates
UPI	Unified Payments Interface — Indian digital payment standard
Surplus	End-of-day leftover food made available to the public at a discounted rate, or completely free
Prep Time	Admin-defined time in minutes required to prepare a menu item
Kitchen Load	Total remaining prep time across all actively preparing orders
Session	A table's active period from when customers are seated to when admin clears the table
Device Fingerprint	A browser-generated identifier stored in localStorage used to track connected devices within a table session
Supabase	Cloud-hosted PostgreSQL database and authentication provider
Vercel	Cloud hosting platform for the Next.js application
Next.js	React-based web framework used for frontend and backend
Tailwind CSS	Utility-first CSS framework used for styling

Table 1: Definitions and Abbreviations

5 Overall Description

5.1 Product Overview

The Restaurant Management System is a web-based application built with Next.js, Supabase, and Tailwind CSS. It serves two types of users simultaneously — internal staff and administrators who manage restaurant operations, and customers who interact with the system through their table's QR code.

The system is designed for single-restaurant deployment. It is not a public platform and is not intended for multi-branch or franchise use.

5.2 User Needs

- Customers need a simple, no-account way to browse the menu and place orders from their table
- Kitchen staff need a clear, real-time view of incoming orders with item-level controls
- Floor staff need visibility into table statuses and order delivery progress
- Admins need full control over the menu, tables, orders, staff accounts, and scheduling
- Restaurant owners need a dashboard that gives them a snapshot of daily operations and customer satisfaction

5.3 Product Context

All users are assumed to be on-premises or connected to the restaurant's network. The system is designed to run on any modern web browser across desktop, tablet, and mobile devices. No installation is required by any user.

6 User Classes and Roles

6.1 Admin

The Admin represents the restaurant owner or manager. Admin accounts can only be created by other admins within the system. Admins have unrestricted access to all features.

- Login and logout via email and password
- Full access to dashboard with operational summary
- Add, edit, delete, and archive menu items
- Manage categories
- Manage tables and generate PIN per session
- View and manage all orders
- Access and interact with kitchen display
- Create, edit, and delete staff and admin accounts
- Edit own profile
- Full control over staff scheduling timetable
- Approve or reject shift switch requests
- Mark end-of-day surplus items and manage public surplus forum
- View customer ratings and feedback on dashboard

6.2 Staff

Staff are restaurant employees including kitchen workers and floor staff. Their accounts are created by an Admin. Staff use the same login but have access to two dedicated views — kitchen and floor.

- Login and logout via email and password
- View full staff scheduling timetable
- Request shift switches with other staff members
- View all menu items and toggle availability
- Access kitchen display — mark individual items as preparing or ready
- View all active orders and update order status
- View all tables and their current statuses
- Create a new session on any table generating a fresh PIN
- Trigger Complete Table when customers are done ordering
- Clear a table session after payment
- View own upcoming shifts

6.3 Customer

Customers are guests of the restaurant. They do not have accounts. They access the system by scanning the QR code on their table and entering the PIN provided by their waiter.

- Scan table QR code and enter PIN to access table session
- Browse available menu items by category
- Add items to cart and place orders
- View real-time status updates for all orders placed from their table
- Add more items during the meal (before payment)
- View auto-generated bill when Admin or Staff triggers Complete Table
- Pay via UPI QR displayed on screen
- Leave optional 1-5 star rating after payment

7 Functional Requirements

7.1 Authentication Module

- FR-A1. The system shall allow Admin and Staff to log in using email and password via Supabase Auth.
- FR-A2. The system shall redirect Admin to the Dashboard upon successful login.
- FR-A3. The system shall redirect Staff to the Orders page upon successful login.
- FR-A4. The system shall display a clear error message when incorrect credentials are entered.
- FR-A5. The system shall allow authenticated users to log out, invalidating their session.
- FR-A6. The system shall prevent unauthenticated users from accessing any Admin or Staff protected pages.
- FR-A7. The system shall fetch the user's role from the profiles table after login and store it in application state.
- FR-A8. Admin shall be able to edit their own profile including name, email, and password.

7.2 Menu Management Module

- FR-M1. The system shall display all non-archived menu items grouped by category.
- FR-M2. Items where `is_available` is false shall appear at the bottom of their category labelled as Not Available Today.
- FR-M3. Archived items shall not appear on the customer-facing menu page.
- FR-M4. Admin shall be able to add a new menu item with name, price, category, and prep time.
- FR-M5. Admin shall be able to edit any existing menu item.
- FR-M6. Admin shall be able to delete any menu item.
- FR-M7. Admin shall be able to archive and unarchive any menu item.
- FR-M8. Admin and Staff shall be able to toggle the daily availability of any menu item.
- FR-M9. Admin shall be able to add, edit, and delete categories.

7.3 Table Management Module

- FR-T1. Admin shall be able to add, edit, and delete tables.
- FR-T2. Staff shall be able to view all tables and their current statuses.
- FR-T3. Each table shall have a unique static QR code that links to that table's session page.

- FR-T4. Admin or Staff shall be able to create a new session on any table, generating a fresh 4-digit PIN.
- FR-T5. Creating a new session on a table that already has an active session shall immediately archive the previous session and disconnect all currently connected devices.
- FR-T6. The table session page shall require the correct PIN before granting access.
- FR-T7. Once the first order is placed from a table session, the PIN entry shall be locked to prevent new devices from joining.
- FR-T8. The system shall track all devices connected to an active session using a browser-generated device fingerprint stored in localStorage.
- FR-T9. Admin shall be able to view the details of any active table including session PIN, connected devices, all orders placed, and running total.
- FR-T10. Staff shall be able to view the basic status of any table.
- FR-T11. The system shall display each table's current status — inactive, open, locked, completed, or cleared.
- FR-T12. All devices that joined a session before it locked shall retain access until the session is cleared.
- FR-T13. Admin or Staff shall be able to trigger Complete Table when customers are done ordering.
- FR-T14. The Complete Table action shall trigger the bill to appear on all connected devices simultaneously via Supabase Realtime.
- FR-T15. After the bill is settled, Admin or Staff shall be able to clear the table session returning it to inactive status.
- FR-T16. Admin shall be able to cancel a session at any time without generating a bill as an emergency clear.
- FR-T17. All completed and cleared sessions shall be archived in the database with timestamps, total amount, device count, and who cleared the session.
- FR-T18. Admin shall be able to view a session history log showing all past sessions per table.

7.4 Order Management Module

- FR-O1. Customers shall be able to add multiple items to a cart from the table session page.
- FR-O2. The system shall automatically calculate and display the cart total.
- FR-O3. Customers shall be able to place an order from their cart.
- FR-O4. All orders placed from the same table session shall be visible to all devices in that session in real time via Supabase Realtime.
- FR-O5. The system shall store the price of each item at time of order as `price_at_order`.

- FR-O6. New orders shall be assigned a status of **placed** automatically.
- FR-O7. Staff and Admin shall be able to update individual item status to **preparing** or **ready** from the kitchen display.
- FR-O8. When all items in an order are marked **ready**, the order status shall automatically update to **ready**.
- FR-O9. Staff and Admin shall be able to mark a **ready** order as **delivered**.
- FR-O10. Orders may be cancelled only before any item is marked as **preparing**.
- FR-O11. No order record shall ever be deleted from the system.
- FR-O12. Admin shall be able to view all orders with table number, item count, timestamp, and status.
- FR-O13. Admin and Staff shall be able to expand any order card to view the full item list with individual item statuses.
- FR-O14. Customers shall be able to add more items to an active session before payment is confirmed.

7.5 Kitchen Display Module

- FR-K1. The system shall provide a dedicated kitchen display view accessible to Staff and Admin.
- FR-K2. The kitchen display shall show all active orders in real time sorted by time placed.
- FR-K3. Each order on the kitchen display shall show individual items with status controls.
- FR-K4. Staff shall be able to mark individual items as **preparing** or **ready** from the kitchen display.
- FR-K5. The kitchen display shall show the estimated wait time for each active order.
- FR-K6. The kitchen display shall update in real time without requiring a page refresh.

7.6 Wait Time Estimation Module

- FR-W1. Admin shall be able to set a prep time in minutes for each menu item.
- FR-W2. When a new order is placed, the system shall calculate the estimated wait time based on the kitchen load.
- FR-W3. Kitchen load shall be calculated as the longest remaining prep time across all items currently in **preparing** status.
- FR-W4. Estimated wait shall be displayed on the customer's table session page after ordering.
- FR-W5. Estimated wait shall be recalculated and updated in real time as items are marked **ready**.

7.7 Billing Module

- FR-B1. The bill shall only be generated when Admin or Staff explicitly triggers the Complete Table action.
- FR-B2. The bill shall aggregate all orders placed across the entire table session, not just a single order.
- FR-B3. The bill shall show all items ordered across all orders, quantities, individual prices, and total amount.
- FR-B4. The bill shall appear simultaneously on all devices connected to the table session via Supabase Realtime.
- FR-B5. The system shall generate a UPI payment QR code with the exact total bill amount pre-filled.
- FR-B6. After the bill is displayed, all connected devices shall show an optional feedback prompt.
- FR-B7. Once the table is cleared by Admin or Staff, the session shall end and all devices shall be logged out.

7.8 Staff Scheduling Module

- FR-SC1. The system shall display a weekly timetable with days on the Y-axis and hours on the X-axis.
- FR-SC2. Each staff member's shift shall be represented as a draggable block on the timetable.
- FR-SC3. Admin shall be able to create new shift blocks by dragging onto the timetable.
- FR-SC4. Admin shall be able to resize shift blocks to change start and end times.
- FR-SC5. Admin shall be able to delete any shift block.
- FR-SC6. All staff shall be able to view the full timetable including all colleagues' shifts.
- FR-SC7. Staff shall be able to click another staff member's shift block and submit a switch request.
- FR-SC8. Switch requests shall only be permitted if the target shift is at least 24 hours away.
- FR-SC9. Admin shall receive a notification when a switch request is submitted.
- FR-SC10. The staff member whose shift is being requested shall also be notified.
- FR-SC11. Admin shall be able to approve or reject any switch request.
- FR-SC12. On approval, the two shift blocks shall be automatically swapped on the timetable.
- FR-SC13. After an approved swap, Admin shall be able to manually adjust any shift blocks to fill gaps.
- FR-SC14. Admin shall be able to manually override shift times regardless of advance notice rules.

7.9 User Management Module

- FR-U1. Admin shall be able to view a list of all Staff and Admin accounts.
- FR-U2. Admin shall be able to create a new Staff or Admin account with name, email, password, and role.
- FR-U3. Admin shall be able to edit the details of any existing user account.
- FR-U4. Admin shall be able to change any user's role between Admin and Staff.
- FR-U5. Admin shall be able to delete any Staff or Admin account.
- FR-U6. A profiles record shall be created automatically whenever a new auth user is registered.
- FR-U7. Admin shall be able to edit their own profile including name, email, and password.

7.10 Dashboard Module

- FR-D1. The Dashboard shall be accessible to Admin only.
- FR-D2. The Dashboard shall display the number of currently active orders.
- FR-D3. The Dashboard shall display the number of tables currently occupied.
- FR-D4. The Dashboard shall display total revenue for today.
- FR-D5. The Dashboard shall display the average customer rating for today.
- FR-D6. The Dashboard shall display the total number of staff accounts.

7.11 Surplus Food Module

- FR-SF1. Admin shall be able to mark menu items as surplus at end of day with a quantity and pickup window.
- FR-SF2. Admin shall be able to set surplus items at a discounted rate, with the flexibility to set the price to zero (100% discount/free) depending on the item.
- FR-SF3. The system shall publish a public surplus forum accessible without login.
- FR-SF4. The surplus forum shall display all currently available surplus items with quantity and pickup window.
- FR-SF5. The surplus forum shall display a running total of meals given away by the restaurant this month.
- FR-SF6. Admin shall be able to mark surplus items as claimed or clear the surplus page.
- FR-SF7. The surplus forum shall automatically hide items whose pickup window has passed.

7.12 Feedback and Ratings Module

FR-FR1. After the bill is displayed, the system shall show an optional 1-5 star rating prompt.

FR-FR2. Customers shall be able to skip the rating prompt without penalty.

FR-FR3. Ratings shall be stored against the order record.

FR-FR4. Admin shall be able to view average ratings and rating breakdown on the Dashboard.

8 Non-Functional Requirements

8.1 Performance

NFR-P1. Page load time shall not exceed 3 seconds under normal network conditions.

NFR-P2. API response time shall not exceed 2 seconds for standard read and write operations.

NFR-P3. The system shall support up to 20 concurrent users without degradation in performance.

NFR-P4. Real-time updates via Supabase Realtime shall reflect within 1 second of the triggering action.

8.2 Usability

NFR-U1. The system shall be fully responsive and usable on desktop, tablet, and mobile screen sizes.

NFR-U2. The system shall be compatible with current versions of Chrome, Firefox, Edge, and Safari.

NFR-U3. Role-based UI differences shall be clearly reflected — unauthorized actions shall not be visible to users who cannot perform them.

NFR-U4. The customer-facing QR page shall be optimized for mobile use.

NFR-U5. Error messages shall be clear and actionable.

NFR-U6. The kitchen display shall be optimized for large screen viewing.

8.3 Security

NFR-S1. All passwords shall be hashed and managed by Supabase Auth.

NFR-S2. Session tokens shall be stored in browser cookies and shall not be accessible via JavaScript.

NFR-S3. All protected API routes shall validate the user's session token and role server-side before executing.

NFR-S4. Row Level Security policies shall be enforced at the database level for all tables.

- NFR-S5. Supabase connection keys shall be stored as environment variables and never hard-coded.
- NFR-S6. The `.env.local` file shall be excluded from version control via `.gitignore`.
- NFR-S7. Table session PINs shall expire when the session is cleared by Admin.
- NFR-S8. Once a table session is locked, new devices shall not be able to join using the same PIN.

8.4 Availability

- NFR-AV1. System availability is dependent on Vercel and Supabase free tier uptime.
- NFR-AV2. No specific uptime guarantee is made beyond what the hosting providers offer by default.

8.5 Maintainability

- NFR-MT1. Code shall be organized according to the folder structure defined in the System Design Document.
- NFR-MT2. Reusable UI elements shall be isolated in the `_components` directory.
- NFR-MT3. Database connection and utility functions shall be isolated in the `_lib` directory.
- NFR-MT4. All code shall be version controlled using Git with meaningful commit messages.
- NFR-MT5. All changes shall be submitted via pull requests and reviewed before merging to main.

9 Use Cases

9.1 UC-01: Customer Places an Order via QR

Field	Detail
Actor	Customer
Precondition	Table session is open. Customer has received PIN from waiter
Main Flow	1. Customer scans QR code on table 2. System displays PIN entry screen 3. Customer enters PIN 4. System grants access to table session page 5. Customer browses menu and adds items to cart 6. Customer places order 7. System creates order with status placed 8. All devices on same session see the order appear in real time 9. System calculates and displays estimated wait time
Alternate Flow	If PIN is incorrect, access is denied and customer is prompted to retry
Postcondition	Order exists in system. Session locks to prevent new devices joining

Table 2: UC-01: Customer Places an Order via QR

9.2 UC-02: Kitchen Staff Updates Item Status

Field	Detail
Actor	Staff or Admin
Precondition	At least one order with status placed exists
Main Flow	1. Staff opens kitchen display 2. System shows all active orders with individual items 3. Staff taps Start on an item 4. Item status updates to preparing, timestamp recorded 5. Estimated wait times recalculate across all pending orders 6. Staff taps Ready when item is done 7. Item status updates to ready 8. When all items in order are ready, order status updates to ready automatically
Postcondition	Order status updated. Customer sees live status change on their device

Table 3: UC-02: Kitchen Staff Updates Item Status

9.3 UC-03: Admin Manages Menu Items

Field	Detail
Actor	Admin
Precondition	Admin is logged in
Main Flow	1. Admin navigates to Menu Management 2. Admin clicks Add Item 3. Admin enters name, price, category, and prep time 4. System creates item with <code>is_available</code> true and <code>is_archived</code> false by default 5. Item appears on menu immediately
Alternate Flow	Admin may edit, delete, archive, or toggle availability of existing items
Postcondition	Menu item created, updated, archived, or deleted in database

Table 4: UC-03: Admin Manages Menu Items

9.4 UC-04: Staff Requests Shift Switch

Field	Detail
Actor	Staff
Precondition	Staff is logged in. Target shift is at least 24 hours away
Main Flow	1. Staff opens scheduling timetable 2. Staff clicks a colleague's shift block 3. Staff clicks Request Switch 4. System validates shift is at least 24 hours away 5. Switch request created with status <code>pending</code> 6. Admin and target staff member are notified 7. Admin reviews and approves the request 8. The two shift blocks are automatically swapped 9. Admin manually adjusts timetable to fill any gaps
Alternate Flow	If shift is less than 24 hours away, request is blocked with an error message
	If Admin rejects, request is closed and no changes are made
Postcondition	Shifts swapped on timetable and visible to all staff

Table 5: UC-04: Staff Requests Shift Switch

9.5 UC-05: Admin Publishes Surplus Food

Field	Detail
Actor	Admin
Precondition	Admin is logged in. End of service approaching
Main Flow	1. Admin navigates to Surplus Management 2. Admin selects menu items to mark as surplus 3. Admin enters quantity and pickup window 4. Admin sets price as discounted, or free (price = 0) 5. System publishes items to public surplus forum 6. Running total of meals given away updates
Alternate Flow	Admin can mark items as claimed or clear the page manually Items past their pickup window are hidden automatically
Postcondition	Surplus items visible on public page until claimed or cleared

Table 6: UC-05: Admin Publishes Surplus Food

9.6 UC-06: Customer Pays and Leaves Rating

Field	Detail
Actor	Customer, Admin or Staff
Precondition	Customers are done ordering. All orders from table delivered
Main Flow	1. Admin or Staff clicks Complete Table on the table management page 2. Bill appears simultaneously on all connected devices 3. Bill shows all items across all orders, quantities, prices, and total 4. UPI QR code displayed with exact total pre-filled 5. Customer scans UPI QR and completes payment externally 6. Admin or Staff clears the table session 7. Rating prompt appears on all connected devices 8. Customers select 1-5 stars or skip 9. Session ends and all devices are logged out 10. Table returns to inactive status
Alternate Flow	Admin can cancel session without bill as emergency clear
Postcondition	Ratings stored. Table inactive and ready for new customers

Table 7: UC-06: Customer Pays and Leaves Rating

10 Constraints and Assumptions

10.1 Constraints

1. The project must be completed within 15 days.
2. The team consists of four members with limited prior experience in React and Next.js.

3. The system is built using Next.js, Supabase, and Tailwind CSS. No alternative frameworks will be considered.
4. The system is deployed on Vercel and Supabase free tier. No paid infrastructure will be used.
5. The system supports a single restaurant only.
6. UPI payment is display-only. The system generates a UPI QR but does not verify or process payment directly.
7. All code changes must be submitted via pull requests and reviewed before merging.
8. Static QR codes are printed per table. Dynamic QR generation per session is not used.

10.2 Assumptions

1. At least one Admin account will be seeded into the system before deployment.
2. All users are assumed to be on-premises or connected to the restaurant's local network.
3. Customers are assumed to have a device with a camera and a modern web browser to scan QR codes.
4. Waiters are responsible for verbally providing the session PIN to customers when they are seated.
5. Menu prep times entered by Admin are assumed to be reasonable estimates, not guaranteed durations.
6. The system assumes stable internet connectivity. Offline mode is not supported.
7. Staff members are assumed to have been briefed on basic system usage before going live.
8. UPI payment is assumed to be completed by the customer independently after scanning the QR.
9. Surplus food pickup is based on trust. The system does not verify whether items were actually collected.